



Git Guide.

Every Git and GitHub answer in one place.

Ask in plain language, or paste the error Git printed.
Exact commands, a danger level for each, and its undo.

an undo for everything · works offline · no trackers · MIT



Amey Thakur

amey-thakur.github.io/GIT-GUIDE



Every Git and GitHub answer in one place.

A note from Amey

Git carries nearly every codebase on earth, and almost everyone who uses it learned it by accident: a command from a teammate, a midnight search, a ritual repeated without understanding.

Git Guide exists to end that way of learning. Ask in plain language, or paste the error Git printed, and get the exact commands with the two things most resources never give: how dangerous each one is, and how to take it back.

These pages hold the whole of it: the model that replaces memorizing, every command section, every error decoded, and the chapters on GitHub and on how teams work. The living version, searchable and always current, waits at **amey-thakur.github.io/GIT-GUIDE**.

Amey



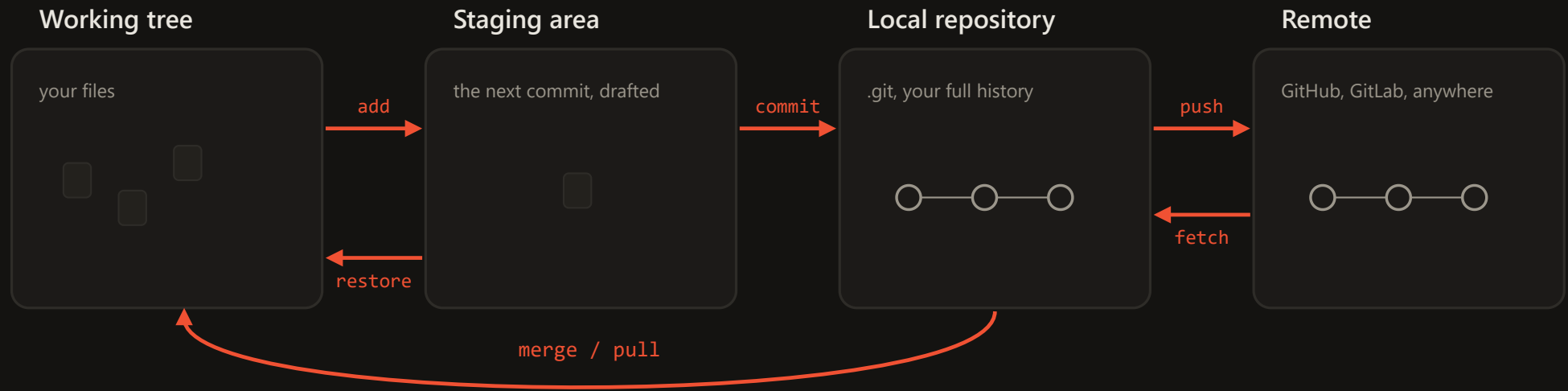
Amey Thakur

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Where your code lives



add copies a change into the staging area, the draft of your next commit. You choose what goes in.

push uploads your commits; **fetch** downloads theirs and stops. **pull** is fetch plus merge.

commit seals the draft into a permanent snapshot in your local repository. Still only on your machine.

restore copies recorded versions back over your files; **reset** moves the branch pointer itself. That is where history is un-happened.



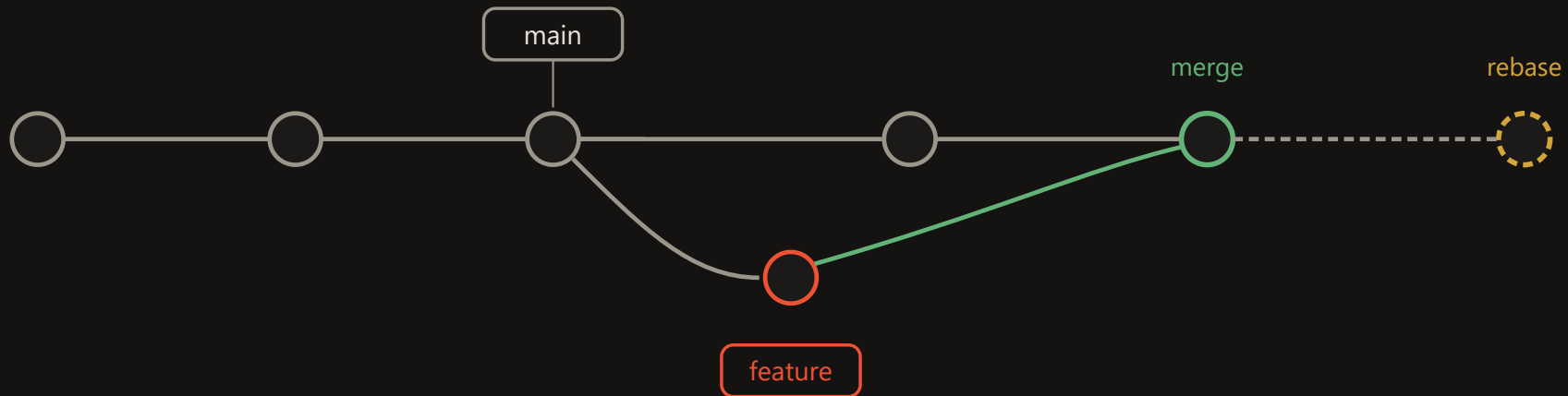
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Every Git and GitHub answer in one place.

Commits, branches, HEAD



A **branch** is a movable label on one commit. Creating one copies nothing; it is instant and free.

A **rebase** replays your commits as new ones for a straight line. It rewrites history: your own unshared work only.

A **merge** commit has two parents and joins the lines. History shows what truly happened; always safe on shared branches.

HEAD is where you stand, normally attached to a branch. Standing on a commit directly is detached HEAD: a state, not an error.



From zero to first push

1 Install

Windows: `winget install Git.Git` · macOS: `brew install git` · Debian: `sudo apt install git`

```
git --version
```

3 One key, every host

Add the public key to GitHub, GitLab, Bitbucket, Azure: the same key works on all

```
ssh-keygen -t ed25519 -C "<email>"
```

5 The loop you will run forever

Edit, stage, commit, push. Everything else is a variation

```
git add <file> && git commit -m "<message>" && git push
```

2 Identity

The same email as your GitHub account, or commits will not count on your profile

```
git config --global user.name "<name>" && git config --global user.email "<email>"
```

4 First repository

Joining: `git clone <url>` · Publishing a folder of yours:

```
gh repo create <name> --public --source . --push
```

6 When it goes wrong

It will, for everyone. Ask the site in plain language, or paste the exact error

```
amey-thakur.github.io/GIT-GUIDE
```



Setup and identity

Once per machine; everything else builds on it.

```
git config --global user.name "<name>"
```

Name stamped into your commits

```
git config --global user.email "<email>"
```

Use your host account email so commits link to your profile

```
git config --global init.defaultBranch main
```

New repositories start on main

```
git config --global core.editor "code --wait"
```

Commit messages open in VS Code

```
git config --global pull.ff only
```

Pull never creates surprise merge commits

```
git config --global fetch.prune true
```

Deleted remote branches disappear locally too

```
git config --global rebase.autoStash true
```

Pull with rebase works even with edits in progress

```
git config --global core.autocrlf true
```

Windows line endings handled once; macOS and Linux use input

```
git config --list --show-origin
```

Every setting and the file it comes from

Start a project

From nothing, or from a URL.

```
git init -b main
```

Turn this folder into a repository

```
git clone <url>
```

Copy a repository, any host, full history

```
git clone --depth 1 <url>
```

Latest snapshot only, fastest

```
git remote add origin <url>
```

Connect a local repository to a host

```
git push -u origin main
```

First push; -u links the branches for good



The daily loop

The eight commands that are most of Git.

```
git status -sb
```

What changed, one line per file

```
git diff
```

Unstaged edits, line by line

```
git diff --staged
```

What the next commit will contain

```
git add <file>
```

Stage a change; git add -p picks piece by piece

```
git commit -m "<message>"
```

Seal the staged snapshot

```
git commit --amend --no-edit
```

Add staged changes to the last commit

```
git fetch
```

See what the remote has before touching anything

```
git pull --rebase
```

Get remote commits, replay yours on top

```
git push
```

Publish your commits

Branches

Cheap, fast, and safe to make by the dozen.

```
git switch -c <branch>
```

Create and move in one step

```
git switch <branch>
```

Move to an existing branch

```
git switch -
```

Back to the branch you just left

```
git branch --show-current
```

Where am I

```
git branch -vv
```

Every local branch with its remote and ahead-behind state

```
git branch -a
```

Every branch, local and remote

```
git push -u origin <branch>
```

Publish a new branch

```
git branch -d <branch>
```

Delete a merged branch; -D forces

```
git push origin --delete <branch>
```

Delete it on the remote too

```
git branch -m <old> <new>
```

Rename



Merge and rebase

Two ways to bring lines together.

```
git merge <branch>
```

Bring a branch in; history stays true

```
git merge --no-ff <branch>
```

Always keep the feature visible as one bubble

```
git rebase main
```

Replay your branch on top of main

```
git rebase -i HEAD~<n>
```

Rewrite, reorder, squash your last n commits

```
git cherry-pick <hash>
```

Copy one commit onto this branch

```
git merge --abort
```

Back out of a conflicted merge

```
git rebase --continue
```

Resume after resolving a conflict

Undo and rescue

Every mistake has a way back.

```
git restore <file>
```

Discard edits, back to last commit

```
git restore --staged <file>
```

Unstage, keep the edits

```
git commit --amend -m "<new message>"
```

Rewrite the last commit message

```
git reset --soft HEAD~1
```

Uncommit, keep everything staged

```
git reset --hard HEAD~1
```

Erase the last commit and its work

```
git revert <hash>
```

Undo by adding an opposite commit; safe when pushed

```
git reflog
```

Every state you have been in; the safety net

```
git reset --hard ORIG_HEAD
```

Undo the merge, rebase, or pull that just happened

```
git clean -nd
```

Preview deleting untracked files; -fd deletes



Stash

A shelf for work that is not ready.

```
git stash push -u -m "<label>"
```

Set everything aside, untracked included

```
git stash list
```

What is shelved

```
git stash pop
```

Bring the newest back and drop it

```
git stash apply stash@{<n>}
```

Bring a specific one back, keep it shelved

```
git stash show -p
```

See inside before applying

Inspect and search

Read history like a book with an index.

```
git log --oneline --graph --all
```

The whole history as a graph

```
git log -p -- <file>
```

Every change to one file; --follow crosses renames

```
git log --grep="<text>"
```

Find commits by message

```
git log -S "<text>"
```

Find commits that added or removed this text

```
git diff main...<branch>
```

What the branch changed since it split; the pull request view

```
git shortlog -sn --all
```

Commits per author, ranked

```
git blame -w <file>
```

Who last touched each line, ignoring whitespace

```
git show <hash>
```

One commit in full; --stat for files only

```
git bisect start
```

Binary-search history for the commit that broke it

```
git branch -a --contains <hash>
```

Which branches carry this commit



Tags and releases

Permanent names on permanent commits.

```
git tag -a v1.0.0 -m "<note>"
```

Mark a release

```
git push origin v1.0.0
```

Tags do not push themselves

```
git tag -d <tag>
```

Delete locally; push origin --delete <tag> for the remote

```
git describe --tags
```

Nearest release to where you stand

```
git switch --detach v1.0.0
```

Visit a release; switch - to come back

Files and ignoring

What Git tracks, and what it must never.

```
printf "node_modules/\n" >> .gitignore
```

Ignore by pattern; affects untracked files only

```
git rm --cached <file>
```

Stop tracking, keep on disk

```
git mv <old> <new>
```

Rename and stage in one step

```
git update-index --skip-worktree <file>
```

Ignore your local edits to a tracked file

```
printf "* text=auto\n" >> .gitattributes
```

Line endings settled for the whole team, committed once

```
touch <folder>/.gitkeep
```

Keep an empty folder



Every platform, same Git

Git is identical everywhere; only the remote URL and sign-in differ. The same SSH public key works on every host.

```
git clone git@github.com:<user>/<repo>.git
```

GitHub over SSH

```
git clone git@gitlab.com:<user>/<repo>.git
```

GitLab

```
git clone git@bitbucket.org:<workspace>/<repo>.git
```

Bitbucket

```
git clone https://dev.azure.com/<org>/<project>/_git/<repo>
```

Azure DevOps (Azure Repos)

```
git clone https://git-codecommit.  
<region>.amazonaws.com/v1/repos/<repo>
```

AWS CodeCommit

```
git config --global credential.helper manager
```

Browser sign-in for HTTPS remotes, all hosts

```
git remote set-url origin <url>
```

Move between hosts or between HTTPS and SSH

```
git clone --mirror <old> && git push --mirror <new>
```

Migrate anywhere with full history

No host at all

Git predates the hosts and needs none of them.

```
git init --bare ~/server/repo.git
```

Your own remote on any machine or shared drive

```
git bundle create repo.bundle --all
```

The whole repository as one file: USB, email, air gap

```
git clone repo.bundle -b main <folder>
```

Clone from the file, no network

```
git format-patch -3
```

Last three commits as mailable patch files

```
git am <file>.patch
```

Apply mailed patches, authorship intact

```
git archive --format=zip -o src.zip HEAD
```

Clean source export, no .git



History surgery

Everything here rewrites history. Fresh clone first, backup kept, force-with-lease after, teammates re-clone.

```
git commit --fixup <hash>
```

Mark a fix as belonging to an earlier commit

```
git rebase -i --autosquash <hash>^
```

Melt fixups into their targets

```
git filter-repo --invert-paths --path <file>
```

Erase a file from all history

```
git filter-repo --strip-blobs-bigger-than 10M
```

Erase every giant blob

```
git push --force-with-lease
```

The only force push worth typing

Scale and speed

Big repositories, kept quick.

```
git clone --filter=blob:none --sparse <url>
```

Huge repository, download almost nothing

```
git sparse-checkout set <folders>
```

Materialize only what you work on

```
git worktree add ../<repo>-review <branch>
```

Second folder, second branch, zero stashing

```
git lfs track "*.psd"
```

Large binaries stored properly

```
git count-objects -vH
```

How heavy is this repository really

```
git maintenance start
```

Background upkeep keeps big repositories fast



Submodules

A repository pinned inside another.

```
git submodule add <url> <path>
```

Pin another repository at an exact commit inside yours

```
git clone --recurse-submodules <url>
```

Clone a repo and its submodules in one step

```
git submodule update --init --recursive
```

Fill the empty folders after a plain clone

```
git submodule update --remote
```

Move pins to each submodule's latest commit

Quality of life

Small settings, daily comfort.

```
git config --global help.autocorrect prompt
```

git statsu asks: run git status instead?

```
git config --global alias.st "status -sb"
```

git st forever

```
git config --global alias.lg "log --oneline --graph --all"
```

git lg: the graph

```
git config --global rerere.enabled true
```

Never resolve the same conflict twice

```
git config --global gpg.format ssh
```

First of three lines to signed, Verified commits

```
git commit --allow-empty -m "rerun"
```

Retrigger CI without touching code

```
git config core.hooksPath .githooks
```

Versioned hooks the whole team shares



How teams run Git

> Working alone

Commit at every working state; push is your backup; branch before experiments

```
git add -A && git commit -m "<message>" && git push
```

> Fork

For repositories you cannot push to, which is all of open source

```
gh repo fork <owner>/<repo> --clone
```

> Releases and hotfixes

A release is a tag; a hotfix is a branch from that tag, merged back after shipping

```
git switch -c hotfix-2.1.1 v2.1.0
```

> Feature branch

The team default: main stays releasable, every change arrives by pull request

```
git switch -c <change> && git push -u origin  
<change>
```

> Trunk-based

Branches live hours, not weeks; tiny changes merge daily behind flags

```
git config --global pull.rebase true
```

The rules that keep teams safe. Never rewrite a branch others build on; rewrite your own freely with force-with-lease. Pull with rebase. Protect main so nothing lands without review and green checks. When anything is unclear: git status names the state you are in, and usually the way out.



Your first repository

Name

Short, lowercase, hyphens: weather-app

Add a README

The front page, and your first commit

Add .gitignore

Pick your language; noise stays out from day one

Choose a license

MIT if unsure; none means all rights reserved

The pull request path

branch · commit · push · open the PR · checks run · review answers with commits · merge, and the branch retires.

`gh pr create --fill`

Open it from the terminal

`gh pr checkout <number>`

Review one locally

`gh pr merge --squash --delete-branch`

Land it clean

The words nobody explains

repository a project and its entire history; everyone says repo

README.md the front page, rendered automatically

Markdown plain text with light marks; GitHub renders it everywhere

.gitignore what Git must never track

LICENSE the legal terms of reuse

public, private everyone reads, only you write; or invitation only

star a bookmark and a thank-you; stars rank repositories

watch subscribe to issues, PRs, and releases

clone vs fork to your machine; to your account

issue a tracked conversation about one bug or request

default branch usually main: what visitors see, what PRs target

gist one shareable file with its own URL

organization a shared account for a team



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

**fatal: not a git repository (or any of the parent directories):
.git**

This folder, and none above it, contains a .git directory. You are outside the project.

remote: Repository not found.

The URL has a typo, the repository is private and you lack access, or you are authenticated as the wrong account.

git@github.com: Permission denied (publickey).

The host never received a key it recognizes: no key loaded, key not added to your account, or an HTTPS problem disguised by an SSH URL.

fatal: Authentication failed for 'https://github.com/...'

The credential your machine sent is wrong, expired, or a password where a token is required.

! [rejected] main -> main (non-fast-forward)

The remote has commits you do not have. Git refuses to overwrite them.

! [rejected] main -> main (fetch first)

Same cause as non-fast-forward: someone pushed before you.

error: src refspec main does not match any

The branch you are pushing does not exist locally, almost always because there is no commit yet.

fatal: refusing to merge unrelated histories

The two sides share no common commit, typically a fresh git init pulled against a remote that already has commits.

CONFLICT (content): Merge conflict in <file>

Both sides changed the same lines; Git needs a human decision.

**error: Your local changes to the following files would be
overwritten by merge**

The operation needs to rewrite files you have edited but not committed. Git refuses to destroy them silently.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

fatal: The current branch has no upstream branch

This local branch has never been connected to a remote branch, so plain git push does not know where to go.

You are in 'detached HEAD' state.

Not an error. You checked out a commit or tag directly, so you stand on a snapshot instead of a branch.

fatal: unable to auto-detect email address

Git has no identity to stamp into the commit.

error: cannot pull with rebase: You have unstaged changes

A rebase must start from a clean tree, and yours has edits.

fatal: unable to access '...': Could not resolve host: github.com

A network problem, not a Git problem: no internet, VPN or proxy in the way, or DNS down.

fatal: unable to access '...': SSL certificate problem

Something intercepts HTTPS, usually a corporate proxy with its own certificate that Git does not trust yet.

error: RPC failed; HTTP 408 curl 22 (or 400, or curl 55)

A large push choked, often against a proxy or server limit on request size.

remote: error: GH001: Large files detected.

GitHub refuses files over 100 MB in normal Git storage.

fatal: Unable to create '.git/index.lock': File exists.

Another Git process is running, or one crashed and left its lock behind.

fatal: detected dubious ownership in repository

The repository folder belongs to a different OS user than the one running Git, common in containers, WSL, and network drives.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

warning: LF will be replaced by CRLF the next time Git touches it

Windows line endings meeting Unix line endings. Harmless, but configure it once and it goes quiet.

error: pathspec '<name>' did not match any file(s) known to git

The file or branch name does not exist as typed: a typo, wrong case, or a branch you have not fetched.

fatal: a branch named '<name>' already exists

You are creating a branch whose name is taken.

error: the branch '<name>' is not fully merged

git branch -d protects commits that exist nowhere else. This is the guardrail working.

fatal: destination path '<folder>' already exists and is not an empty directory

git clone creates the folder, and a folder with that name is already there.

error: remote origin already exists

You are adding a remote named origin to a repository that already has one.

fatal: 'origin' does not appear to be a git repository

You pushed or pulled against a remote name that is not configured here.

Your branch and 'origin/main' have diverged

Both you and the remote gained different commits since you last synced.

Your branch is ahead of 'origin/main' by N commits

Not an error. You have local commits the remote lacks; push when ready.

fatal: bad object HEAD (or: loose object is corrupt)

The object database is damaged, usually a crash, a full disk, or a synced-folder service fighting over .git.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

error: unable to create file <path>: Filename too long

Windows' 260-character path limit, hit by deep folder structures.

error: invalid path '<name>'

The repository contains a file name Windows forbids: reserved names like aux or con, or characters like colon and question mark, committed from Linux or macOS.

warning: You appear to have cloned an empty repository.

Not an error. The remote exists but has no commits yet.

fatal: this operation must be run in a work tree

You are inside the .git folder itself, or in a bare repository that has no working files by design.

fatal: tag '<name>' already exists

Tags are unique names; this one is taken.

error: cannot spawn .git/hooks/pre-commit: No such file or directory

A hook script is missing, not executable, or has Windows line endings breaking its shebang line.

error: gpg failed to sign the data

Signing is switched on but the signing setup is incomplete or the key is unavailable.

error: Pulling is not possible because you have unmerged files.

An earlier merge stopped at conflicts and is still waiting for you to finish it.

hint: You have divergent branches and need to specify how to reconcile them.

Newer Git asks you to choose a pull strategy once instead of silently merging.

fatal: could not read Username for 'https://github.com': No such device or address

Git wanted to ask for credentials but no terminal was attached, the classic CI and script failure.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

`xcrun: error: invalid active developer path
(/Library/Developer/CommandLineTools)`

A macOS update removed the command line tools, and git on a Mac lives inside them.

`Could not open a connection to your authentication agent.`

ssh-add needs the ssh-agent process, and none is running in this shell.

`fatal: Not possible to fast-forward, aborting.`

Your pull is set to fast-forward only, and the branches have diverged, so a decision is needed.

`You are not currently on a branch.`

Pull needs a branch to update, and you stand on a detached commit.

`error: failed to push some refs to '<url>'`

The wrapper line under most push rejections; the real reason is the line above it, usually non-fast-forward.

`fatal: could not create work tree dir '<name>': Permission denied`

You are cloning into a folder your user cannot write to.

`error: insufficient permission for adding an object to repository database .git/objects`

Files inside .git belong to another user, typically after running Git with sudo once or sharing a repo folder.

`error: cannot lock ref 'refs/remotes/origin/<branch>': unable to resolve reference`

A stale or corrupted remote-tracking reference, often after an interrupted fetch.

`error: cannot lock ref 'refs/heads/<x>/<y>': 'refs/heads/<x>' exists`

Branch names are paths: a branch named x blocks any branch named x/y, and the reverse.

`fatal: bad revision 'HEAD'`

The repository has no commits yet, so HEAD points at nothing.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

fatal: ambiguous argument 'HEAD': unknown revision or path not in the working tree.

Same family: an empty repository, or a name Git cannot tell apart from a file.

error: The following untracked working tree files would be overwritten by checkout

The branch you are switching to tracks files that exist here untracked; Git refuses to destroy them silently.

error: unable to unlink old '<file>': Permission denied

Something holds the file open: an editor, a running process, or an antivirus scan, most often on Windows.

fatal: '<branch>' is already checked out at '<path>'

Worktrees guarantee one checkout per branch, and that branch lives in another folder.

error: You need to resolve your current index first

A merge stopped at conflicts, and Git will not move on until you finish or abandon it.

error: commit <hash> is a merge but no -m option was given.

Reverting a merge needs to know which parent line survives.

fatal: bad config line <n> in file <path>

A typo in a config file: an unclosed quote, a stray character, or a section header gone wrong.

There is no tracking information for the current branch.

Pull does not know which remote branch feeds this one.

fatal: couldn't find remote ref <branch>

No branch by that name exists on the remote: a typo, a deleted branch, or main versus master.

Host key verification failed.

SSH does not yet trust this server's fingerprint, or the stored one no longer matches.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

ssh: connect to host github.com port 22: Connection timed out
The network blocks port 22, the classic corporate and campus firewall move.

error: RPC failed; curl 92 HTTP/2 stream 0 was not closed cleanly

A proxy or middlebox mishandling HTTP/2 during a large transfer.

fatal: the remote end hung up unexpectedly

The connection died mid-transfer, and Git may report it as early EOF. Usually a large clone or push meeting a proxy, server, or network limit.

warning: adding embedded git repository: <folder>

You added a folder that is itself a Git repository; it would be recorded as an empty pointer, confusing every clone.

fatal: in unpopulated submodule '<path>'

You are standing inside a submodule folder that was never initialized, an empty shell.

fatal: no submodule mapping found in .gitmodules for path '<path>'

A submodule folder is recorded in the index but missing from .gitmodules, a half-removed submodule.

error: could not apply <hash>... <message>

A rebase or cherry-pick hit a conflict on this commit and paused for you.

CONFLICT (modify/delete): <file> deleted in <ref> and modified in <ref>

One side deleted the file, the other edited it; Git cannot choose.

remote: Support for password authentication was removed on August 13, 2021.

GitHub no longer accepts account passwords over HTTPS; a token or SSH key is required.

remote: Write access to repository not granted.

Your account can read this repository but not push to it.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

fatal: Not a valid object name: 'main'.

Branching from main before any commit exists; a branch cannot point at nothing.

remote: Permission to <repo> denied to github-actions[bot].

A workflow tried to push, but Actions runs with read-only permissions unless the workflow asks for more.

fatal: your current branch 'main' does not have any commits yet

Log, branch, and friends need at least one commit to talk about.

fatal: shallow update not allowed

You are pushing from a shallow clone, common in CI checkouts.

fatal: Refusing to fetch into current branch

refs/heads/<branch> of non-bare repository

Fetching directly into the branch you are standing on would pull the floor from under you.

**error: Server does not allow request for unadvertised object
<hash>**

A submodule is pinned to a commit the server will not hand out directly, or a shallow clone wants what it cannot see.

fatal: 'origin/<branch>' is not a commit and a branch '<name>' cannot be created from it

Your clone has never heard of that remote branch; the start point does not exist locally.

error: branch '<name>' not found.

No local branch by that exact name; a typo, or it lives only on the remote.

fatal: There is no merge to abort (MERGE_HEAD missing).

Nothing is mid-merge; whatever state you are in, it is not that.

fatal: No rebase in progress?

The rebase you are trying to continue or abort already ended, or never started.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

**fatal: could not read Username for 'https://github.com':
terminal prompts disabled**

GIT_TERMINAL_PROMPT=0 is set, as in most CI, and Git wanted to ask for credentials.

remote: HTTP Basic: Access denied

GitLab's phrasing of bad HTTPS credentials: an expired or wrong token, or a password where a token belongs.

gpg: signing failed: No secret key

Signing is on, but the configured key is not on this machine, typical after a laptop change.

gpg: signing failed: Inappropriate ioctl for device

GPG wants to ask for your passphrase but has no terminal to ask in.

fatal: Could not read from remote repository.

The generic tail of every SSH failure; the true cause sits in the lines above it.

**fatal: unable to access '<url>': OpenSSL SSL_read: Connection
was reset, errno 10054**

Something on the network path, a proxy, VPN, or security software, cut the TLS connection, most often on Windows.

**fatal: unable to access '<url>': error setting certificate
verify locations**

Git's CA bundle path is broken, usually after a partial install or an overzealous cleanup.

error: bad signature 0x00000000 (fatal: index file corrupt)

The index file was damaged, a crash, a full disk, or a sync service writing into .git.

fatal: LF would be replaced by CRLF in <file>

The strict form of the line-ending check: safecrlf is set to refuse converting changes instead of warning.

fatal: It seems that there is already a rebase-merge directory

A previous rebase never finished; its state directory still sits in .git.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

fatal: unsafe repository ('<path>' is owned by someone else)

The older phrasing of the ownership guard, same cause as dubious ownership.

! [remote rejected] <branch> -> <branch> (pre-receive hook declined)

The server ran a policy check and said no: a protected branch, an oversized file, or a failed lint, and it printed its reason just above this line.

error: GH013: Repository rule violations found for refs/heads/<branch>.

GitHub rulesets blocked the push; the lines above name the exact rule, commonly a required pull request or signed commits.

error: GH006: Protected branch update failed for refs/heads/<branch>.

That branch only accepts changes through reviewed pull requests.

remote: error: File <name> is <size> MB; this exceeds GitHub's file size limit of 100.00 MB

One specific file crossed the hard limit; the push cannot land while any commit carries it.

husky - pre-commit hook exited with code 1 (error)

The project's own pre-commit checks failed: lint, tests, or formatting, and the real output sits just above.

Everything up-to-date

Not an error: push found nothing to send. The surprise usually means the work is uncommitted or on another branch.

nothing to commit, working tree clean

Commit found no staged changes. Either everything is already committed, or the edits live elsewhere.

The requested URL returned error: 403

The server recognized you and still said no: wrong account's cached credentials, or no write permission here.

fatal: cannot do a partial commit during a merge.

You named specific paths while a merge is in progress. A merge commit must record the whole resolved state, not a slice of it.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

fatal: You have not concluded your merge (MERGE_HEAD exists).

A previous merge stopped with conflicts and was never finished or aborted. Git will not start anything new on top of it.

error: Entry '<file>' not uptodate. Cannot merge.

That file has uncommitted edits, and the merge needs to change it. Git refuses rather than overwrite them.

fatal: Exiting because of an unresolved conflict.

A command that needed a clean tree found conflict markers still in place from an earlier operation.

warning: refname 'main' is ambiguous.

A branch and a tag share the same name, so Git cannot tell which one you mean.

git: 'comit' is not a git command. See 'git --help'.

A typo, or a subcommand that comes from a tool you have not installed, like lfs or filter-repo.

fatal: protocol 'https' is not supported

The URL carries an invisible character, almost always from copying a link out of a web page or chat.

fatal: Unable to find remote helper for 'https'

This Git build has no HTTPS transport, typically a minimal install or one missing its curl support.

Warning: you are leaving 2 commits behind, not connected to any of your branches:

You committed while in detached HEAD and are now switching away. Those commits have no branch pointing at them.

fatal: Needed a single revision

Git could not resolve the reference you gave, commonly a stash entry, branch, or hash that does not exist.

error: object file .git/objects/ab/cdef... is empty

An object was truncated, usually by a crash, a full disk, or a sync service writing inside .git.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

`fatal: transport 'file' not allowed`

Git blocks submodules and clones from local file paths by default, a hardening change after a security advisory.

`remote: fatal: pack exceeds maximum allowed size`

The push bundles more data than the server accepts in one go, which GitHub caps at 2 GB.

`fatal: Out of memory, malloc failed (tried to allocate ... bytes)`

Packing a large repository exceeded available memory, often on a small CI runner or container.

`error: unable to create symlink <path>: Function not implemented`

The repository contains symbolic links and this system will not create them, typically Windows without permission or a filesystem that lacks support.

`warning: unable to rmdir '<path>': Directory not empty`

Git removed everything it tracked in that folder, but something untracked remains, most often a leftover submodule or build output.

`Encountered 1 file(s) that should have been pointers, but weren't:`

Files matching an LFS rule were committed as ordinary blobs, usually because they went in before LFS was installed or tracked.

`batch response: This repository is over its data quota.`

LFS storage or bandwidth for the account is exhausted. Bandwidth is consumed by every clone and fetch of LFS content, not only by uploads.

`gh: Not Found (HTTP 404)`

The resource does not exist, or your token cannot see it. Private repositories return 404 rather than 403 so their existence stays hidden.

`gh: To use GitHub CLI in a GitHub Actions workflow, set the GH_TOKEN environment variable.`

gh is installed on the runner but has no credentials; workflows do not inherit a login.

`remote: error: GH007: Your push would publish a private email address.`

Your commits carry a real address while your account has email privacy enabled, so GitHub blocks the push.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

The process `'/usr/bin/git'` failed with exit code 128

A Git command inside the workflow failed and Actions reports only the exit code. The cause is nearly always credentials or a missing ref.

Error: Input required and not supplied: token

An action expected an input that resolved to an empty string, usually a secret that does not exist under that name.

fatal: update_ref failed for ref 'refs/heads/main': cannot lock ref

Another Git process holds the reference, or a stale lock file was left behind by one that crashed.

The previous cherry-pick is now empty, possibly due to conflict resolution.

Every change in that commit already exists on this branch, so applying it produces nothing.

No changes - did you forget to use 'git add'?

A rebase or cherry-pick paused for you to resolve something, and you ran continue without staging the result.

hint: Waiting for your editor to close the file...

Git opened your configured editor for a commit message and is paused until that program exits. Nothing is stuck.

modified: <path> (modified content, untracked content)

Not an error. The submodule folder contains uncommitted or untracked work of its own.

warning: unable to access '/root/.gitconfig': Permission denied

Git is running as a user whose home directory it cannot read, common inside containers and CI.

fatal: empty ident name (for <you@example.com>) not allowed

user.name is set to an empty string, so Git cannot build a valid author line.

warning: redirecting to https://github.com/<owner>/<repo>.git/

The remote URL still uses http, or omits the .git suffix, so the server redirects every request.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

fatal: packed object <hash> (stored in .git/objects/pack/...) is corrupt

A pack file was damaged, usually by disk trouble, an interrupted transfer, or a sync client.

warning: The last gc run reported the following. Please correct the root cause and remove .git/gc.log

Background maintenance failed, often because too many unreachable objects are held by the reflog or another process was running.

warning: Clock skew detected. Your build may be incomplete.

Commit timestamps sit in the future relative to this machine, usually a wrong system clock or files restored from another timezone.

fatal: Invalid refsPEC '<value>'

The branch or refsPEC argument is malformed, often an empty variable in a script or a stray colon.

fatal: reference is not a tree: <hash>

The commit you named does not exist in this clone, commonly after a force push removed it or in a shallow clone that never had it.

fatal: repository '<url>' does not exist (during submodule update)

The URL in .gitmodules points somewhere that moved, was renamed, or is private to you.

error: cannot 'squash' without a previous commit

The first line of an interactive rebase list is marked squash or fixup, and there is nothing above it to fold into.

error: invalid line 1: pcik <hash>

A typo in the interactive rebase instruction list. Git refuses to guess at an unknown command word.

fatal: --unshallow on a complete repository does not make sense

This clone already has full history, so there is nothing to deepen. Usually a CI script assumes a shallow checkout that was not shallow.

CONFLICT (rename/delete): <file> renamed in HEAD but deleted in <branch>

One side renamed a file while the other removed it, so Git cannot tell whether the result should exist.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

error: Could not verify signature. No principal matched.

The commit is signed with an SSH key that is not listed in the allowed-signers file, so Git cannot vouch for who made it.

! [rejected] v1.0.0 -> v1.0.0 (already exists)

That tag is already on the remote, and Git will not silently move a published tag.

Cannot rewrite branches: You have unstaged changes.

History-rewriting tools refuse to run against a dirty working tree, because the rewrite would fight your edits.

fatal: expected flush after ref listing

Something other than Git answered the connection: a proxy, a captive portal, or a server error page.

pull request create failed: GraphQL: No commits between main and feature

The branch has nothing the base does not already have, so there is nothing to review.

gh: could not determine base repository

gh works out the repository from the folder's remotes, and this folder has none it recognizes.

Invalid workflow file: .github/workflows/ci.yml#L12

The YAML does not parse or breaks the workflow schema. Indentation and a missing colon cause most of it.

**error: external filter 'git-lfs filter-process' failed fatal:
<file>: smudge filter lfs failed**

Checkout tried to expand an LFS pointer and the LFS binary was missing, unauthenticated, or out of quota.

##[error]Process completed with exit code 1.

Not a Git problem. A command in the step returned non-zero, and the runner reports the step as failed.

Error: Container action is only supported on Linux

A Docker-based action was used in a job running on a Windows or macOS runner.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

Error: Failed to create deployment (status: 404) with build version <sha>

The deploy step ran before Pages was switched to the Actions source, so there is nothing to deploy into.

error: chmod on .git/config.lock failed: Operation not permitted

The filesystem will not let Git set permissions: typically a network share, an exFAT drive, or a Windows folder reached from WSL.

warning: ignoring broken ref refs/remotes/origin/HEAD

The pointer recording the remote's default branch is empty or malformed, often after the default branch was renamed.

kex_exchange_identification: Connection closed by remote host

The SSH handshake was cut before authentication: usually a firewall, a captive network, or rate limiting on port 22.

WARNING: UNPROTECTED PRIVATE KEY FILE! Permissions 0644 for '<path>' are too open.

SSH refuses to use a private key that other users on the machine can read.

Load key "<path>": invalid format

The key file is damaged, in a format OpenSSH does not read, or you pointed at the public key by mistake.

sign_and_send_pubkey: signing failed: agent refused operation

The agent holds the key but will not sign with it, usually because of a permissions problem on the key or a confused agent session.

error: empty filename in tree entry fatal: unable to read tree <hash>

A commit object was written with a malformed entry, usually by a third-party tool or a damaged import.

Automatic merge failed; fix conflicts and then commit the result.

Both sides changed the same lines, so Git stopped and left the file with both versions marked for you to choose.

error: key does not contain a section: user

A config key was written without its section, most often user instead of user.name.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

fatal: --local can only be used inside a git repository

A repository-scoped config command ran in a folder that is not a repository, common in scripts and fresh shells.

HTTP 422: Validation Failed (a pull request already exists for <owner>:<branch>)

That branch already has an open pull request against the same base. GitHub allows only one.

**npm ERR! Error while executing: git ls-remote -h -t
ssh://git@github.com/<owner>/<repo>.git npm ERR! exited with
error code: 128**

A dependency points at a Git URL and npm shells out to Git, which has no key for that host in this environment.

**ERROR: Cannot find command 'git' - do you have 'git' installed
and in your PATH?**

A requirement uses a git+ URL, and the container or runner has no Git binary.

**Make sure you configure your 'user.name' and 'user.email' in
git.**

VS Code refuses to commit without an identity, and its terminal may not see the shell profile where you set one.

**nothing added to commit but untracked files present (use "git
add" to track)**

The files are new, so Git sees them but is not tracking them yet. Commit records only tracked, staged content.

**Error: Unable to resolve action `actions/checkout@v9`,
repository or version not found**

The action reference names a tag, branch, or repository that does not exist. A typo in the version is the usual cause.

**The head commit for this pull_request event is not ahead of the
base commit.**

The branch has nothing the base lacks, so the merge-result checkout is empty and the workflow stops.

**dependabot.yml: The property '#/updates/0' did not contain a
required property of 'schedule'**

Every entry under updates needs package-ecosystem, directory, and schedule. One is missing.

error: could not lock config file .git/config: File exists

A stale .git/config.lock is left over from a crashed Git process or an editor holding the file open.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

error: cannot run vi: No such file or directory error: unable to start editor 'vi'

core.editor points at a program this machine does not have, common in minimal containers.

error: Cannot delete branch 'main' checked out at '<path>'

Git refuses to delete the branch you are standing on, or one checked out in another worktree.

fatal: <name> - not something we can merge

The branch name does not resolve: a typo, or a remote branch you have not fetched.

warning: Cannot merge binary files: <file> (HEAD vs. <branch>)

Binary files cannot be merged line by line, so Git leaves one side in place and asks you to choose.

Unable to push: file is locked by another user

Git LFS file locking is enabled and someone else holds the lock on that path.

! [rejected] v1.0 -> v1.0 (would clobber existing tag)

A fetch would move a tag you already have, which Git refuses without permission.

gh: this API operation needs the "read:org" scope

The stored token lacks a scope the command needs.

Warning: Failed to restore: Cache service responded with 503

The Actions cache backend was briefly unavailable. The job continues without a cache.

Page build failed: Your site is having problems building

The Pages build errored, usually a Jekyll parse failure or an unsupported plugin.

The custom domain <domain> is already taken

Another GitHub repository or account already claims that domain.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

fatal: 'feature branch' is not a valid branch name

Branch names cannot contain spaces, and reject `..`, `~`, `^`, `:`, and trailing dots.

warning: more than one credential helper configured

Several helpers are set across config levels, and they can answer with different accounts.

remote: Permission to <owner>/<repo>.git denied to <other-username>.

Git authenticated as a different account than the one with access: usually a cached credential.

Hi <other-user>! You've successfully authenticated, but GitHub does not provide shell access.

Not an error, but the name is the wrong account: SSH offered a different key than you intended.

Nothing to do (interactive rebase exited immediately)

The instruction list was saved empty or every line was commented out, so the rebase had nothing to run.

error: a cherry-pick or revert is already in progress hint: try "git cherry-pick (--continue | --abort | --quit)"

An earlier pick stopped on a conflict and was never finished.

fatal: unable to access '<url>': gnutls_handshake() failed: Error in the pull function.

The TLS handshake failed, typically an intercepting proxy or an outdated Git build.

fatal: unable to access '<url>': Empty reply from server

Something closed the connection without answering: a proxy, a firewall, or a captive portal.

fatal: protocol error: bad line length character: <chars>

The server sent something Git could not parse, usually an HTML error page or an SSH login banner.

fatal: cannot fork() for index-pack: Resource temporarily unavailable

The machine hit a process or memory limit while unpacking the transfer.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

error: refs/heads/<name> does not point to a valid object!

The ref file names an object this repository no longer holds.

error: unable to read sha1 file of <path> (<hash>)

An object the index references is missing from the object store.

warning: multiple merge bases found, using the first

History has more than one common ancestor, usually after criss-cross merges.

fatal: <path>: '<path>' is outside repository at '<root>'

The path argument resolves above the repository root.

The following paths are ignored by one of your .gitignore files:

You asked Git to add a file that an ignore rule excludes.

error: Sparse checkout leaves no entry on working directory

The sparse patterns match nothing, so the checkout would be empty.

Aborting commit due to empty commit message.

The editor closed with no text, so Git refused to record the commit.

no changes added to commit (use "git add" and/or "git commit -a")

Files are modified but nothing is staged, so there is nothing to record.

Your name and email address were configured automatically based on your username and hostname.

Git guessed an identity from the machine, and commits made this way are attributed to nobody.

fatal: invalid date format: <value>

The date string does not match any format Git understands.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

error: dst refspec <name> matches more than one

A tag and a branch share that name, so the push target is ambiguous.

error: unable to delete '<branch>': remote ref does not exist

The branch is already gone from the remote, or the name is misspelled.

fatal: No configured push destination.

This branch has no remote to push to, and none was given on the command line.

You have exceeded a secondary rate limit. Please wait a few minutes before you try again.

Too many writes in a short window. This limit is separate from the hourly quota.

Error: Resource not accessible by integration

The workflow token lacks permission for what the step attempted.

At least 1 approving review is required by reviewers with write access.

A ruleset requires approvals this pull request does not have.

Required status check "build" is expected.

A required check never reported, so the pull request waits indefinitely.

This branch has conflicts that must be resolved

The pull request cannot merge until the overlap with its base branch is settled.

This repository has been archived by the owner. It is now read-only.

Archived repositories reject pushes, issues, and pull requests by design.

Your SSO session has expired or you must re-authenticate to access this organization.

A SAML-protected organization needs the credential or session authorized again.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

No space left on device

The runner filled its disk, usually with container layers or build output.

Error: Unable to find any artifacts for the associated workflow run

The download ran for an artifact that was never uploaded, or belongs to a different run.

The job was canceled because "ubuntu-latest" failed

One matrix combination failed and fail-fast cancelled the others.

This scheduled workflow is disabled because the repository has been inactive for 60 days

Scheduled runs pause automatically after a long quiet period.

error parsing called workflow: workflow was not found

The reusable workflow path or ref does not resolve for the caller.

npm ERR! code EINTEGRITY npm ERR! integrity checksum failed when using sha512

The lockfile records a hash that no longer matches what the registry served.

ERROR: Command errored out with exit status 128: git clone -q <url>

pip shelled out to Git and Git failed, almost always on authentication.

verifying <module>: checksum mismatch SECURITY ERROR

The module checksum differs from the recorded sum, which can mean a moved tag or a poisoned cache.

failed to solve: failed to read dockerfile: error fetching default branch

Docker tried to build from a Git URL it could not read.

Failed to execute git clone --mirror -- '<url>'

Composer's Git call failed, usually credentials for a private package.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

Git not found. Install Git, or configure git.path in your settings.

VS Code cannot locate a Git binary on PATH.

Push rejected: Push to origin/main was rejected

The IDE is reporting a non-fast-forward push behind a friendlier dialog.

Agreeing to the Xcode/iOS license requires admin privileges

macOS ships Git through the developer tools, which refuse to run until the licence is accepted.

error: unable to create file <name>: Invalid argument

Windows reserves names such as aux, con, and nul, so a checkout containing one fails.

warning: the following paths have collided (e.g. 'README.md' and 'readme.md')

Two paths differ only by case, which a case-insensitive filesystem cannot represent at once.

fatal: '<path>' is a locked working tree

The worktree was locked, usually because it lives on removable media.

fatal: validation failed, cannot remove working tree: gitdir file does not exist

The worktree folder was deleted without unregistering it.

error: no note found for object <hash>

That commit carries no note in the current notes ref.

fatal: You need to start by "git bisect start"

A bisect subcommand ran outside a bisect session.

fatal: Some good revs are not ancestors of the bad rev

The good commit is not on the path leading to the bad one, so bisect cannot search between them.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

error: patch failed: <file>:12 error: <file>: patch does not apply

The patch expects surrounding lines that no longer match.

fatal: previous rebase directory .git/rebase-apply still exists

An earlier git am or rebase stopped and was never finished.

fatal: No names found, cannot describe anything.

The repository has no tags reachable from this commit.

error: gc is already running on machine '<host>' pid <n>

Another maintenance process holds the lock, or one crashed and left it.

error: Cannot prune, other worktrees may be using objects

Pruning is unsafe while another worktree may reference the objects.

Aborting: Refusing to destructively overwrite repo history since this does not look like a fresh clone

filter-repo insists on a fresh clone so a mistake cannot destroy your only copy.

fatal: A git directory for '<name>' is found locally with remote(s)

A previous submodule of that name left its git directory behind in .git/modules.

error: Submodule '<path>' contains modified content

The submodule has uncommitted work, so the parent refuses to record a clean pointer.

warning: remote HEAD refers to nonexistent ref, unable to checkout

The remote's default branch pointer names a branch that no longer exists.

remote: error: GH008: Your push referenced at least <n> unknown Git LFS objects

The commits reference LFS objects the server never received.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

warning: commit-graph file is corrupt, ignoring it

The cached commit graph is damaged; Git falls back to reading history directly.

fatal: unsupported index version <n>

An older Git is reading an index written by a newer one.

fatal: exceeded maximum include depth (10)

Config includes form a loop, so Git stops before recursing forever.

fatal: cannot run <path>/pre-commit: No such file or directory
core.hooksPath points somewhere that does not exist, or the hook is not executable.

hooks/pre-commit: /bin/sh^M: bad interpreter: No such file or directory

The hook script has Windows line endings, so the shebang line is unreadable.

husky - command not found

Hooks reference husky but dependencies were never installed in this checkout.

files were modified by this hook

A formatting hook rewrote files, so the staged content no longer matches.

gh: command not found

GitHub CLI is not installed or not on PATH.

extension not found: gh-<name>

The extension is not installed for this account or machine.

could not find any workflows named <name>

The workflow file does not exist on the branch gh is looking at.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

Error: Unable to get ACTIONS_ID_TOKEN_REQUEST_URL env variable

The job requested an OIDC token without permission to mint one.

Error: Unable to write to \$GITHUB_ENV

The step tried to write a runner file that does not exist in this context.

Error: Get Pages site failed. Please verify that the repository has Pages enabled

The deploy ran before Pages was switched on for the repository.

Error: Artifact exceeds maximum size for Pages deployment

The uploaded site is larger than the Pages limit.

You have exceeded your Codespaces usage limit

The monthly included hours or storage are spent.

denied: installation not allowed to Write organization package

The token lacks package write permission for that organization.

unauthorized: unauthenticated: User cannot be authenticated with the token provided.

The registry login used a token without the right scope, or an expired one.

npm ERR! 403 Forbidden - PUT https://registry.npmjs.org/<pkg>

The account may not publish that package name, or the version already exists.

HTTPError: 400 Bad Request File already exists

That exact version was already uploaded, and PyPI never allows a replacement.

GitLab: You are not allowed to push code to protected branches on this project.

The branch is protected on GitLab, the same idea as a GitHub ruleset.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

`remote: Invalid credentials fatal: Authentication failed for 'https://bitbucket.org/...'`

Bitbucket rejected the password: it requires an app password, not an account password.

`! [remote rejected] refs/pull/1/head -> refs/pull/1/head (deny updating a hidden ref)`

A mirror push tried to write pull request refs, which servers keep read-only.

`fatal: unable to checkout working tree warning: Clone succeeded, but checkout failed.`

The objects arrived but the files could not be written: usually a path or permission problem.

`warning: <path>: symlink not created, symlinks are not supported`

The filesystem or platform will not create symbolic links, so Git wrote a plain file.

`fatal: bad numeric config value '<value>' for '<key>': invalid unit`

A size or count setting was given a value Git cannot parse. Units are k, m, and g.

`error: invalid key: <key>`

The key is malformed, usually a missing section or a stray character.

`fatal: transport 'ftp' not supported`

The remote URL uses a protocol this Git build cannot speak.

`error: unknown option '<flag>'`

The flag does not exist for that command, or belongs to a different subcommand.

`fatal: -L requires a range and a file`

Line-range logging needs both the range and the path, joined by a colon.

`fatal: Invalid pathspec magic in ':(exclude)'`

The pathspec syntax is wrong or unquoted, so the shell mangled it.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

fatal: cannot describe '<hash>' (no annotated tags)

describe reads annotated tags by default, and only lightweight ones exist.

fatal: no tag message?

An annotated tag was requested without a message and the editor closed empty.

No local changes to save

There is nothing modified or staged, so a stash would hold nothing.

fatal: '<ref>' is not a stash-like commit

The reference given is not a stash entry.

fatal: clean.requireForce defaults to true and neither -i, -n, nor -f given

Git refuses to delete untracked files without an explicit instruction.

error: the following file has local modifications: (use --cached to keep the file, or -f to force removal)

Removing a file with uncommitted edits would discard them.

fatal: destination exists, source=<a>, destination=

The target path already exists, so the move would overwrite it.

fatal: could not resolve --source=<ref>

The commit, branch, or tag named as the source does not resolve.

fatal: a branch is expected, got commit '<hash>'

switch works on branches; a bare commit needs an explicit detach.

fatal: this worktree is not sparse

A sparse-checkout command ran in a repository where the feature is off.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

warning: rerere.enabled is not set; nothing recorded

Conflict resolutions are only remembered when the feature is switched on.

error: Repository lacks these prerequisite commits:

The bundle is incremental and this repository lacks the commits it builds on.

git: 'send-email' is not a git command.

send-email ships separately from core Git on most platforms.

**error: Load key "<path>": No such file or directory fatal:
failed to write commit object**

Signing is configured with a key file that does not exist.

git: 'credential-manager' is not a git command

The configured credential helper is not installed on this machine.

error: unable to read askpass response from '<program>'

Git tried to prompt through a helper program that failed or is missing.

**fatal: unable to access '<url>': error:1408F10B:SSL
routines:ssl3_get_record:wrong version number**

Something is speaking plain HTTP to an HTTPS port, usually a misconfigured proxy.

remote: error: Commits must have verified signatures.

A ruleset requires signed commits and one in the push is unsigned.

HTTP 409: Conflict

The resource changed since you read it, or the repository is empty.

HTTP 410: Issues are disabled for this repo

The feature was switched off for the repository.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

HTTP 401: Bad credentials

The token is wrong, revoked, or expired.

Invalid workflow file: Unrecognized named-value: 'env'

An expression referenced a context that is not available where it was used.

The workflow is not valid: job 'a' and job 'b' are in a dependency cycle

Two jobs list each other in needs, which can never resolve.

Failed to initialize container <image>: health check failed

A service container never became ready, so the job aborted.

Error: This version of actions/download-artifact is not compatible with artifacts uploaded by v3

Artifact versions must match between upload and download.

Cache not found for input keys: <key>

No cache matches that key, which is normal on the first run or after a lockfile change.

Waiting for approval from required reviewers

The job targets a protected environment and is paused until someone approves.

CodeQL did not detect any code written in languages supported by CodeQL

The scan found nothing to analyse, usually a build that produced no compiled output.

Dependabot cannot update <package> to a non-vulnerable version

No fixed release exists yet, or your constraints forbid the one that does.

An error occurred setting up the container: failed to build

The devcontainer image failed to build, so the environment never started.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

Domain does not resolve to the GitHub Pages server

The DNS record is missing, wrong, or has not propagated yet.

Certificate not yet issued for <domain>

The certificate is still being provisioned after a DNS change.

HTTP 422: Validation Failed asset with this name already exists

That filename is already attached to the release.

release not found

The release does not exist, or the tag was never pushed.

Could not resolve to a ProjectV2 with the number <n>

The project number belongs to another owner, or the token lacks project scope.

Cannot add issue to a closed milestone

Closed milestones do not accept new items.

The repository <name> already exists on this account.

That name is taken by another repository you own, possibly archived.

You don't have admin rights to transfer this repository

Transfers require admin on the source and permission to create in the target.

User is not provisioned in this organization

An identity-managed organization only admits users its provider has provisioned.

warning: <path> has different normalization forms

macOS stores certain accented names differently, so the same name can appear twice.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

warning: commit message did not conform to UTF-8

The message contains bytes that are not valid UTF-8, usually from a terminal in another codepage.

error: cannot rebase: Your index contains uncommitted changes.

Rebase rewrites commits and refuses to run with staged work in the way.

warning: 3 lines add whitespace errors.

The patch introduces trailing whitespace or space-before-tab.

error: promisor remote not found

A partial clone lost the remote it fetches missing objects from.

fatal: unknown value for config 'protocol.version': 2

The server or an intermediary does not speak protocol v2.

fatal: <branch> is already used by worktree at <path>

One branch cannot be checked out in two worktrees at once.

error: Untracked working tree file '<file>' would be overwritten by merge. (applying a stash)

The stash carries an untracked file you have since recreated.

HEAD detached at <hash> (inside a submodule)

Submodules are pinned to a commit, so they always check out detached.

Skipping object checkout, Git LFS is not installed.

Pointer files were checked out literally because the filters are missing.

no pull requests found for branch "<branch>"

gh looked for a pull request from the current branch and found none.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

GraphQL: Could not resolve to a Repository with the name '`<owner>/<repo>`'

The repository does not exist, was renamed, or is invisible to this account.

X could not create alias: "`<name>`" is already an alias

That alias name is taken.

fatal: alias loop detected: expansion of '`<name>`' does not terminate

An alias expands to itself, directly or through a chain.

fatal: cannot handle `<text>` as a builtin

The alias runs shell syntax without being marked as a shell alias.

error: There was a problem with the editor '`<editor>`'.

The editor exited with a failure status, so Git discarded the message.

WARNING: terminal is not fully functional

The TERM value is unknown, so the pager cannot draw properly.

error: cannot run `<pager>`: No such file or directory (custom diff pager)

A diff pager such as delta is configured but not installed here.

This message is displayed because '`merge.tool`' is not configured.

No merge tool is set, so Git cannot launch one.

Launch '`<tool>`' [Y/n]?

difftool asks before each file, which is tedious across a large diff.

fatal: no such path '`<path>`' in HEAD

The path does not exist in the commit you are blaming, usually after a rename or delete.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

fatal: --follow requires exactly one pathspec

Rename following works for one path at a time.

bisect run failed: exit code 128 from '<command>' is < 0 or >= 128

The test script returned a status bisect reserves for its own signalling.

git grep found no matches (exit status 1)

Not an error: grep signals absence with a non-zero status, which stops strict scripts.

fatal: prefix must end with /

git archive requires the prefix to be a directory path.

fatal: unable to update url base from redirection

The server redirected to a different repository path than the one requested.

fatal: clone of '<url>' into submodule path '<path>' failed

A submodule clone failed, usually authentication or a moved URL.

error: atomic push failed, no refs updated

One ref in an atomic push was rejected, so all of them were refused together.

warning: There are too many unreachable loose objects; run 'git prune' to remove them.

Rewrites and resets left many orphaned objects behind.

old mode 100644 new mode 100755

Only the executable bit changed, which is real to Git but usually accidental.

The following paths are ignored (a negation cannot re-include inside an excluded directory)

Once a directory is excluded, Git never descends into it, so a negation inside cannot match.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

warning: core.excludesFile points to a file that does not exist
The global ignore file is configured but missing.

error: external filter '<name>' is not available
A .gitattributes filter names a program this machine does not have.

warning: commit <hash> has a future date
A commit timestamp is ahead of the current clock, so ordering looks wrong.

error: unable to write file <path>: Permission denied (Windows, file in use)
Another process holds the file open, commonly an antivirus scanner or an editor.

error: unable to unlink old '<file>': Invalid argument (inside a synced folder)
A cloud sync client is holding or rewriting files while Git works on them.

Visual Studio: Git failed with a fatal error. Authentication failed for '<url>'
Visual Studio reuses the Windows credential store, and the entry there is stale.

Exception caught during execution of add command
org.eclipse.jgit.api.errors.JGitInternalException
The IDE's own Git implementation hit a locked index, usually alongside a command-line operation.

SourceTree: The embedded Git version is out of date
The bundled Git is older than the repository features in use.

Android Studio: No version control systems are configured for this project
The IDE has not been told the project folder is a repository.

Xcode: Source control is disabled for this project
Xcode's source control integration is switched off, or the repository is outside the project root.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

`hudson.plugins.git.GitException: Failed to fetch from <url>`

The Jenkins agent could not authenticate or reach the remote.

`Permission denied (publickey). Please make sure you have the correct access rights (CircleCI)`

The CI project has no valid checkout key for the repository.

`fatal: could not read Username (Azure Pipelines checkout)`

The pipeline checkout has no credential for a second repository.

`Failed to download module: error downloading '<url>':
/usr/bin/git exited with 128`

Terraform shelled out to Git for a module source and Git failed.

`fatal: [host]: FAILED! => {"msg": "Failed to checkout branch
<name>"}`

The Ansible git module could not check out that reference on the target host.

`error: failed to authenticate when downloading repository`

`attempted ssh-agent authentication`

Cargo used SSH and found no usable key in the agent.

`Bundler::GitError: Git error: command `git fetch` in directory
<path> has failed.`

Bundler's cached Git source could not update.

`[ERROR] Failed to execute goal ... scm:checkout Cannot
checkout: <url>`

The Maven build tried a Git checkout it could not authenticate.

`Unable to load the service index for source <url> 401
(Unauthorized)`

The package feed rejected the credential, commonly a GitHub Packages token without read:packages.

`error: failed to resolve dependencies Couldn't get the list of
tags`

Swift Package Manager could not read the repository's tags, usually access or a moved URL.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

[!] Error installing <pod> fatal: could not read Username for 'https://github.com'

CocoaPods fetched a private pod source with no credential available.

Error: Cannot determine base for affected commands: fatal: bad revision 'origin/main'

The monorepo tool compared against a ref the shallow CI checkout never fetched.

No new changesets found (release workflow exits without publishing)

The release tool found no changeset files, so there is nothing to version.

There is no previous release, the next release version is 1.0.0

Not an error: the tool found no prior tag and is starting the version history.

pre-commit not found. Did you forget to activate your virtualenv?

The hook script calls a tool missing from this environment.

No staged files match any configured task

Nothing staged matches the configured patterns, so the tool had nothing to run.

subject may not be empty [subject-empty] type may not be empty [type-empty]

The message does not match the enforced convention, usually missing the type prefix.

git-crypt: error: file not encrypted: <path>

An encrypted-path file was committed before git-crypt was unlocked.

Failed to get the data key required to decrypt the SOPS file

No key available here can decrypt the secrets file.

Aborting: refusing to run on a repo with a working tree

BFG expects a bare mirror clone, not a working checkout. BFG is a separate download that needs Java.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

Author: <user> not defined in authors file

The SVN import found a username with no mapping to a Git identity.

hg-fast-export.sh: command not found

The Mercurial conversion script is not installed or not on PATH.

codespace not found or not running

The codespace is stopped or was deleted.

Error: Failed to install feature '<feature>'

A devcontainer feature failed during the image build.

This job is waiting for a runner from '<label>' to come online

No runner with that label is available to take the job.

Canceling since a higher priority waiting request for '<group>' exists

A newer run in the same concurrency group superseded this one.

The job was not started because recent account payments have failed

Actions is blocked because billing is unpaid or the spending limit is reached.

fatal: repository '../<name>.git' does not exist

The submodule uses a relative URL that resolves against a different parent than expected.

Resolving deltas: 100% (done) but the command never returns

Delta resolution finished and Git is now writing files, which is disk-bound on huge trees.

Updating files: 100% (12000/12000), done. (taking minutes)

Every file write is being scanned, which multiplies checkout time on Windows.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

warning: Cannot merge binary files (custom merge driver not defined)

A .gitattributes entry names a merge driver that is not configured here.

CHANGELOG.md merged with duplicated entries

A union merge keeps both sides, which duplicates lines that were meant to be one.

warning: disabling cone pattern matching (non-cone patterns present)

Non-cone sparse patterns disable the fast sparse index.

jq: error: null has no keys

The API returned no data for that path, so the filter had nothing to read.

only 30 results returned

The endpoint paginates and the request took only the first page.

We couldn't deliver this payload: Service Timeout

The receiving endpoint did not answer inside the delivery window.

Resource not accessible by personal access token

The token type cannot reach that resource; some endpoints require a GitHub App installation token.

Dependabot could not authenticate to the private registry

Dependabot has its own secret store, separate from Actions secrets.

Secret <name> is not available in this codespace

Codespaces secrets are granted per repository and must be selected for this one.

You must enable two-factor authentication to access this organization

The organization requires two-factor authentication and your account has none.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

fatal: --orphan is incompatible with -b

Those two ways of creating a branch cannot be combined.

fatal: Cannot force update the current branch.

A branch cannot be moved with -f while it is checked out.

fatal: Invalid branch name: '<name>'

The new branch name breaks the naming rules, often a leading dash or a space.

fatal: Refusing to fetch into branch 'refs/heads/main' checked out at '<path>'

Fetching directly into a checked-out branch would desynchronise the working tree.

! [rejected] <name> -> <name> (fetch first)

The remote branch moved after you last fetched.

warning: reflog of 'HEAD' references pruned commits

The reflog names commits that garbage collection has already removed.

dangling blob <hash>

An object exists that nothing references, usually staged-then-reset content.

**broken link from commit <a> to tree **

A commit points at a tree the repository no longer holds.

error: index-pack died of signal 9

The unpack process was killed, almost always by the out-of-memory killer.

sign_and_send_pubkey: no mutual signature algorithm

The client and server share no acceptable signature algorithm, typical with an old RSA key.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

Received disconnect from <host>: Too many authentication failures

The agent offered too many keys before the right one.

WARNING: REMOTE HOST IDENTIFICATION HAS CHANGED!

The server's host key differs from the one recorded, which SSH treats as serious.

Unverified: The signing key is not known to GitHub

The commit is signed, but the public key is not on the account.

Unverified: This commit was created without a signature

Vigilant mode marks every unsigned commit, including old ones.

You have triggered an abuse detection mechanism.

The request pattern looked automated and aggressive.

refusing to allow a Personal Access Token to create or update workflow without `workflow` scope

Pushing changes to files under .github/workflows needs an extra scope.

Error: The operation was canceled. (job exceeded timeout-minutes)

The job ran past its allowed time and was stopped.

The <image> runner image is deprecated and will be removed

The runner label points at an image being retired.

Error: Unable to find Node version '<version>' for platform linux

That exact runtime version is not available for the runner platform.

Warning: Cache save failed: Unable to reserve cache with key <key>, another job may be creating this cache

Two jobs raced to save the same cache key.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

Error: Artifact name is not valid: <name>. Contains the following character: Double quote

Artifact names cannot contain characters that are illegal in file paths.

Error: The environment url is not a valid URL

The deployment URL output was empty or malformed.

Error: Deployment request failed: another deployment is in progress

Two Pages deployments overlapped.

Release is a draft and cannot be downloaded

Draft releases are visible only to maintainers and serve no assets publicly.

Some labels could not be transferred

Labels that do not exist in the target repository are dropped during a transfer.

CODEOWNERS errors: <line> is not a valid owner

An owner entry is misspelled, not a member, or lacks repository access.

Changes must be made through a pull request.

A ruleset blocks direct pushes to that branch, including for administrators.

Pull request was removed from the merge queue

The pull request failed its checks against the queued result, so the queue ejected it.

Auto-merge is not enabled for this repository

The repository setting that allows auto-merge is switched off.

Squash merging is not allowed on this repository

The repository permits only the merge methods its settings enable.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

Making this repository private will detach all public forks
Public forks cannot follow a repository into private visibility.

The repository name you entered does not match
Deletion requires typing the full owner and repository name exactly.

Import failed: unable to access the source repository
The importer could not read the source, usually credentials or an unsupported protocol.

remote: warning: File <name> is 63.00 MB; this is larger than GitHub's recommended maximum file size of 50.00 MB
A warning, not a rejection: the push succeeds but the repository is heading somewhere painful.

Rewriting history will break every existing clone and fork
A force-push after a rewrite leaves everyone else on incompatible history.

Nothing specified, nothing added. hint: Maybe you wanted to say 'git add .'?
git add ran with no paths, so it had nothing to stage.

no changes added to commit Untracked files: <file>
Untracked files are never committed implicitly, not even with commit -a.

fatal: You have nothing to amend.
There is no previous commit on this branch to amend.

fatal: Failed to resolve 'HEAD' as a valid ref.
The branch has no commits, so HEAD points at nothing to reset to.

error: short object ID <hash> is ambiguous
The abbreviated hash matches more than one object.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

fatal: invalid reference: <branch>

The branch named for the new worktree does not exist.

fatal: <hash>: cannot cherry-pick a root commit without --allow-empty

The commit has no parent, so there is no diff to apply.

warning: skipped previously applied commit <hash>

The commit's change is already present upstream, so the rebase dropped it.

fatal: Could not find merge strategy '<name>'.

The strategy name is misspelled or unavailable in this Git version.

error: while searching for: error: patch fragment without header at line <n>

The patch file is truncated or was mangled by copy and paste.

5.7.8 Username and Password not accepted

The SMTP server rejected the credentials, commonly because an app password is required.

Automatic notes merge failed. Fix conflicts in .git/NOTES_MERGE_WORKTREE

Two sides added notes to the same object.

warning: object <hash> is replaced by <other>

A replace ref is rewriting history on the fly, which can be baffling if you did not create it.

fatal: No web server found

git instaweb needs a local web server it could not locate.

fatal: operation not supported by protocol (git archive --remote)

The server does not allow remote archiving.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

**fatal: remote error: You can't push to
git://github.com/<owner>/<repo>.git**
The git protocol is read-only; pushes need HTTPS or SSH.

fatal: repository '<url>' not found (plain file server)
The URL serves files but is not a Git endpoint.

fatal: unable to connect to cache daemon
The credential cache helper is configured but its daemon is not running.

error: storing password in keychain failed
The macOS keychain is locked or denied access to Git.

remote: Invalid username or password. (after rotating a token)
The credential store still holds the previous token.

warning: <path> is a git repository, but not a submodule
A nested repository exists where a submodule is expected.

**This repository is configured for Git LFS but 'git-lfs' was not
found on your path**
A clone carries LFS configuration on a machine where LFS is missing.

Error: working copy must not be dirty
LFS migration rewrites history and refuses to run over uncommitted work.

Error: Could not parse mailmap line
The mailmap file used during a rewrite has a malformed entry.

fatal: gc is already running (during a fetch)
Background maintenance is holding the repository while another command wants it.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

warning: commit-graph has incorrect fanout value

The cached graph disagrees with the objects, usually after an interrupted write.

error: multi-pack-index checksum does not match

The multi-pack index is damaged and must be regenerated.

warning: core.fsmonitor is set but the daemon is unavailable

The filesystem monitor is enabled but cannot run in this environment.

warning: untracked cache is disabled on this system or location

The filesystem does not report directory changes reliably enough to cache them.

The pattern does not match because of a trailing space

Trailing spaces are significant in .gitignore unless escaped.

The attribute has no effect on files already committed

Attribute changes apply on the next checkout or after renormalising.

error: unable to write file <path>: Text file busy

The file is currently executing, so it cannot be replaced.

error: unable to write sha1 file: Disk quota exceeded

The account or volume ran out of space mid-operation.

error: unable to create directory for <path>: Read-only file system

The checkout location cannot be written to.

The rename was not recorded because only the case changed

A case-insensitive filesystem reports no change, so Git records nothing.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

Reinitialized existing Git repository in <path>

Not an error: git init ran where a repository already exists and only refreshed it.

fatal: not a git repository: '<path>' (GIT_DIR is set)

A stale GIT_DIR or GIT_WORK_TREE variable is pointing Git somewhere else.

error: unknown option '--recurse-submodules'

The installed Git predates the flag you used.

fatal: repository '<url>/info/refs' not found

The server is not exposing the Git HTTP endpoints at that path.

fatal: the receiving end does not support push options

The server does not accept the -o push options you sent.

The value of the GH_TOKEN environment variable is being used for authentication.

A token in the environment overrides the account you logged in as.

Error: Required property is missing: shell

Every run step in a composite action must declare its shell.

Error: Cannot find entrypoint <path> in the container

The Docker action's entrypoint is missing or not executable.

Invalid workflow file: while scanning for the next token found character '\t' that cannot start any token

YAML forbids tabs for indentation.

The job output is empty in the dependent job

Outputs must be declared on the job, not only written by the step.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

Error: Matrix must define at least one vector

The matrix resolved to nothing, usually from an expression returning an empty list.

Error: Secret <name> is not available in the called workflow

Reusable workflows receive no secrets unless they are passed.

Dependabot could not create a pull request: resource not accessible

Dependabot runs with its own restricted permissions, separate from Actions.

CodeQL analysis failed: out of memory

The analysis exceeded the runner's memory on a large codebase.

remote: error: Push cannot contain secrets remote: GITHUB PUSH PROTECTION

Push protection detected a credential in the commits and refused the push.

gitleaks: leaks found: 1

The local scanner matched something credential-shaped in the staged content.

Liquid Exception: Liquid syntax error in <file>

Jekyll tried to parse template syntax in a file that is not a template.

404 on every asset when the site is served under a subpath

Absolute paths break when a project site lives at /<repo>/ rather than the domain root.

No release notes were generated

Automatic notes come from merged pull requests, and none were found in the range.

failed to upload release asset: context deadline exceeded

The upload exceeded the time limit, usually a large file on a slow link.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

You have locked conversation on this issue

Locking blocks new comments, including your own until unlocked.

This category does not support marking answers

Only Q&A categories have accepted answers.

fatal: repository '<url>.wiki.git' not found

The wiki has never been created, so its repository does not exist yet.

The README does not appear on my profile

The special repository must be named exactly like your username and be public.

My contributions are not visible on my profile

Private contributions are hidden unless you opt in.

The achievement did not appear after merging a pull request

Achievements are computed on a delay and only count qualifying activity.

This invitation is no longer valid

Organization invitations expire, typically after a week.

Cannot add member: user is managed by an external identity provider

Team membership is controlled upstream by the identity provider.

Access denied: your IP address is not permitted

An enterprise IP allowlist blocks the address you are connecting from.

The dependency graph is empty for this repository

No supported manifest was detected, so there is nothing to inventory.



The errors you will meet

The message, then the cause. Every fix is one search away **on the site**.

`verification failed: no matching attestations found`

The artifact has no attestation from that repository, or it was built elsewhere.

`Updates were rejected because a pushed branch tip is behind its remote (mirror)`

The mirror target has commits the source lacks, so mirroring would lose them.

`This repository has been archived and is read-only (push rejected)`

Archived repositories accept no pushes until unarchived.

`The new repository has no commit history from the template`

Templates deliberately start with a single fresh commit.

`The forwarded port is private and cannot be shared`

Forwarded ports default to private visibility.

`postCreateCommand failed with exit code 1`

The setup command failed, so the environment started incomplete.

`your branch is based on 'origin/<name>', but the upstream is gone.`

The remote branch was deleted, usually after its pull request merged.

`fatal: You are in the middle of a rebase -- cannot push`

Git refuses to publish a branch that is mid-rewrite.

`You are currently bisecting. (committing mid-session)`

Committing during a bisect confuses the search and leaves stray commits.

`fatal: cannot stash during a merge (unmerged paths present)`

Stashing cannot represent a half-resolved merge.



The living version

Eight doors, one address: amey-thakur.github.io/GIT-GUIDE

Finder

ask anything, or paste
the error

Start

zero to first push

Learn

the model, step by step

Fix

guided rescue

Errors

Git's messages,
decoded

GitHub

first repo to branch
protection

Workflows

how teams run Git

Cheatsheet

everything, one line
each

SAFE

REWRITES HISTORY

DESTRUCTIVE

every answer carries its danger level and its undo

Every answer also lives in the repository's Discussions; a question the guide cannot answer becomes its next addition. MIT licensed, no trackers, works offline.



Amey Thakur

amey-thakur.github.io/GIT-GUIDE



Every Git and GitHub answer in one place.